

S1988

Vintage Sampler Emulation

Capture, Stretch, Loop & Resample



RAVE generation

USER MANUAL

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1. Introduction

S1988 is a vintage sampler emulation plugin inspired by late-1980s 16-bit hardware samplers. This plugin captures the distinctive character of early digital sampling while adding modern creative capabilities including granular time-stretching, resonant filtering, and lo-fi degradation effects.

Those machines were known for a warm, slightly gritty character that came from its 16-bit converters, analog filters, and the inherent limitations of early digital audio. S1988 recreates these characteristics while providing intuitive controls for everything from subtle vintage coloring to extreme sound mangling.

Whether you're adding retro character to drums and samples, creating experimental soundscapes with time-stretching, or designing lo-fi textures, S1988 delivers that coveted late-80s sampler sound with modern flexibility.

2. Key Features

2.1 Vintage Sampler Character

- 18 dB/octave resonant low-pass filter modeled after classic sampler topology
- 18 dB/octave high-pass filter for frequency shaping
- Warm preamp saturation with musical gain compensation
- Filter drive for additional harmonic content

2.2 Lo-Fi Processing

- Variable sample rate reduction (4000 Hz - 44100 Hz)
- Bit depth reduction (4-bit to 16-bit) for digital degradation
- DC blocking to maintain clean output even with extreme settings

2.3 Time Stretch Engine

- Granular time-stretching with variable grain size (2-200 ms)
- Time factor from 25% to 400% for dramatic tempo manipulation
- Pitch Comp switches Time Factor between time stretching and pitch shifting
- Transpose (± 24 semitones): pitch and speed together, like a varispeed
- Reverse playback mode for creative effects

2.4 Flexible Output

- Equal-power dry/wet mix for parallel processing
- Auto-gain compensation to maintain consistent output levels
- Mono/Stereo processing modes
- Zero latency operation

2.5 Tape Loop

- Capture Last: loop the audio you just heard, straight away
- Loop Speed from -200% to +200%: slow down, stop and reverse like tape
- Overdub with Feedback for layered loops that fade over time
- Resample: print what you hear back into the loop, generation after generation
- Warble: random tape wow
- Loop Order: loop the stretched sound, or stretch the held loop

3. User Interface

The S1988 interface is organized into logical sections that follow the signal flow. Controls are grouped by function for intuitive operation:

- Input Section: Input level and preamp controls
- Filter Section: High-pass, low-pass filters with resonance and drive
- Lo-Fi Section: Sample rate and bit depth reduction
- Time Stretch Section: Granular time manipulation controls
- Loop Section: capture, loop length, speed, overdub and Resample; ORDER sits in its title bar
- Tape Section: Warble
- Output Section: Mix, output level, and processing mode

Controls that do nothing in the current state are dimmed but stay adjustable: the stretch controls while Stretch is off, the loop controls while Loop is off, and Loop Length while Loop Sync is not on Free. Hover over a control to see what it does on the hint line at the bottom; the i button beside it switches the hints off and on.

4. Input Section

The input section provides gain staging and saturation controls that shape the signal before filtering.

Parameter	Range	Default	Description
Input	-20 to +20 dB	0 dB	Input gain control. Sets the level entering the processor.
Preamp	0 - 100%	0%	Warm saturation stage. Adds harmonic content and soft clipping with automatic gain compensation.
Auto Gain	Off / On	Off	Automatic output level compensation based on preamp, filter drive, and resonance settings.

5. Filter Section

The filter section provides 18 dB/octave filters with resonance and drive capabilities. These filters respond immediately to parameter changes for dynamic, responsive filtering.

5.1 Filter Parameters

Parameter	Range	Default	Description
HP Filter	20 - 2000 Hz	20 Hz	High-pass filter cutoff frequency. 18 dB/octave slope removes low frequencies.
LP Filter	200 - 20000 Hz	20000 Hz	Low-pass filter cutoff frequency. 18 dB/octave slope with resonance.
Resonance	0 - 100%	0%	Filter resonance (Q). Higher values create emphasis at the cutoff frequency.
Filter Drive	0 - 100%	0%	Post-filter saturation. Adds grit and edge to the filtered signal.

5.2 Filter Topology

The S1988 uses cascaded filter stages to achieve its 18 dB/octave slopes:

- Low-Pass Filter: 12 dB biquad with resonance + 6 dB one-pole cascade
- High-Pass Filter: 6 dB one-pole + 12 dB biquad cascade

This topology matches the behavior of classic hardware samplers while maintaining stability at high resonance settings.

6. Lo-Fi Section

The Lo-Fi section recreates the digital degradation characteristics of vintage samplers with variable sample rate and bit depth reduction.

Parameter	Range	Default	Description
Sample Rate	4000 - 44100 Hz	44100 Hz	Reduces the effective sample rate. Lower values create aliasing and digital artifacts.
Bit Depth	4 - 16 bits	16 bits	Reduces bit resolution. Lower values create quantization noise and stepped waveforms.

6.1 Classic Settings Reference

Hardware Era	Sample Rate	Bit Depth	Character
late-80s hardware sampler (1988)	44100 Hz	16-bit	Clean, warm
Early Samplers (1985)	22050 Hz	12-bit	Gritty, classic
8-bit Era (1980s)	11025 Hz	8-bit	Crunchy, lo-fi
Extreme Lo-Fi	4000 Hz	4-bit	Destroyed

6.2 Tape

Parameter	Range	Default	Description
Warble	0 - 100%	0%	Random tape wow: a slow pitch drift of up to about half a semitone. 0% is off.

7. Time Stretch Section

The time stretch engine uses granular synthesis to manipulate time and pitch independently. This section enables creative effects from subtle time adjustments to extreme sonic manipulation.

Parameter	Range	Default	Description
Stretch	Off / On	Off	Enables the time stretch engine. When off, audio passes through normally.
Time Factor	25 - 400%	100%	With Pitch Comp off: above 100% the audio plays slower, below 100% faster, at the same pitch (200% = twice as long). With Pitch Comp on: shifts the pitch instead (200% = one octave up).
Grain Size	2 - 200 ms	50 ms	Size of granular windows. Smaller = smoother transients, larger = more artifacts.

Parameter	Range	Default	Description
Pitch Comp	Off / On	On	Sets what Time Factor changes. Off: the timing, at the same pitch. On: the pitch, at the same timing.
Transpose	-24 to +24 st	0 st	Pitch and speed together, in semitones, like a varispeed. Works while Stretch is on.
Reverse	Off / On	Off	Plays the stretch and the loop backwards.

7.1 Grain Size Guidelines

- 2-10 ms: Very smooth, minimal artifacts, best for subtle stretching
- 20-50 ms: Balanced, good for most material
- 50-100 ms: Audible granular artifacts, creative textures
- 100-200 ms: Extreme granular effects, rhythmic stuttering

7.2 Stretching Without Jumps

On a running track, a stretch that plays slower, faster or backwards (Time Factor with Pitch Comp off, Transpose, Reverse) jumps from time to time to stay with the track. To stretch a phrase without jumps, loop it first: switch Loop on and set Loop Order to Loop>Stretch. Press Resample to print the stretched loop.

8. Loop Section

The loop section enables real-time audio looping with tempo synchronization capabilities. When enabled, incoming audio is captured and played back in a continuous loop, allowing you to freeze and manipulate audio material.

Parameter	Range	Default	Description
Loop	Off / On	Off	Enables loop capture and playback. When turned on, audio is captured and looped continuously.
Loop Sync	Free / 1/1 / 1/2 / 1/4 / 1/8 / 1/16 / 1/32 / 1/64 / 1/128	Free	Loop timing mode. Free uses milliseconds; other values sync to host tempo using note divisions.
Loop Length	10 - 4000 ms	500 ms	Loop duration in milliseconds when Loop Sync is set to Free. Ignored when using tempo-synced divisions.
Capture	Next / Last	Next	Next records the coming loop length, then loops it. Last loops the audio you just heard, straight away.
Loop Speed	-200 to +200%	100%	Tape speed of the held loop. The pitch follows the speed; through 0% the loop stops and plays backwards.
Overdub	Off / On	Off	Records the input onto the held loop. You hear it live, then on every pass.

Feedback	0 - 100%	100%	How much of the loop survives each Overdub pass. 100% keeps every layer; lower values let older layers fade.
Resample	Button	Off	Press while Loop is on: what you just heard, with every effect on it, becomes the new loop. Each press adds another generation.
Loop Order	Stretch>Loop / Loop>Stretch	Stretch>Loop	The ORDER pill in the LOOP title bar. Stretch>Loop loops the stretched sound. Loop>Stretch puts the loop first, so the stretch works on the held loop live.

8.1 Loop Behavior

With Capture on Next, switching Loop on records the coming loop length, then playback begins. With Capture on Last, it loops the audio you just heard, straight away. The loop plays until Loop is turned off.

With Loop Order on Stretch>Loop (the default) the loop holds the stretched sound, so the stretch settings shape what gets captured. Set Loop Order to Loop>Stretch to put the loop first: Time Factor, Grain Size and Transpose then work on the held loop live.

8.2 Tempo Sync

When Loop Sync is set to a note division (1/1 through 1/128), the loop length is calculated from the host DAW tempo. For example, at 120 BPM with Loop Sync set to 1/4, the loop will be exactly one beat (500 ms). This ensures your loops stay perfectly synchronized with your project.

8.3 Overdub and Resample

Overdub records onto the held loop while you play; Feedback sets how fast older layers fade.

Resample prints the sound you hear into the loop, effects included, and the knobs then act on top of it again. Turn an effect up, press Resample, turn it back down: the effect stays in the loop. The dry signal from Mix is not printed.

Tape loop, overdub and warble are based on chompi, which includes a limiter from stmlib. Both are used under the mit license.

9. Output Section

The output section provides final gain staging and processing mode options.

Parameter	Range	Default	Description
Mix	0 - 100%	100%	Dry/wet blend using equal-power crossfade. 0% = dry only, 100% = wet only.
Output	-20 to +20 dB	0 dB	Final output level with safety limiting to ± 1.0 .
Processing	Mono / Stereo	Stereo	Processing mode. Mono sums L+R before output.

10. Signal Flow

Understanding the signal flow helps optimize your settings and troubleshoot any issues:

Input → Input Gain → Stretch Buffer → Time Stretch → Loop →

Preamp → HP Filter → LP Filter → Filter Drive →

Sample Rate Reduction → Bit Reduction → DC Block → Warble →

Mono/Stereo → Auto Gain → Mix → Output

10.1 Key Signal Flow Notes

- Loop Order sets where the loop sits: after the time stretch (default) or before it
- Time stretch is processed first, so filters respond to stretched audio
- Preamp saturation feeds into filters, creating harmonic interactions
- Filter drive is post-filter for raw edge on clean-filtered signal
- Lo-fi processing occurs after filtering to preserve filter character
- DC blocker removes offset from bit crushing artifacts
- Mix blending uses equal-power curves for musical parallel processing
- Resample prints the wet signal, before Mix, into the loop

11. Complete Parameter Reference

This section provides a complete reference of all automatable parameters in S1988.

Parameter	Range	Default	Notes
Input	-20 to +20 dB	0 dB	Input gain
Preamp	0 - 100%	0%	Warm saturation
Auto Gain	Off / On	Off	Level compensation
HP Filter	20 - 2000 Hz	20 Hz	18 dB/oct HPF
LP Filter	200 - 20000 Hz	20000 Hz	18 dB/oct LPF
Resonance	0 - 100%	0%	Filter Q
Filter Drive	0 - 100%	0%	Post-filter saturation
Sample Rate	4000 - 44100 Hz	44100 Hz	Rate reduction
Bit Depth	4 - 16 bits	16 bits	Bit reduction
Stretch	Off / On	Off	Enable stretch
Time Factor	25 - 400%	100%	Stretch ratio
Grain Size	2 - 200 ms	50 ms	Granular window
Pitch Comp	Off / On	On	Time Factor: time or pitch
Transpose	-24 to +24 st	0 st	Pitch and speed together
Reverse	Off / On	Off	Reverse playback
Loop	Off / On	Off	Enable loop
Loop Sync	Free / 1/1 - 1/128	Free	Loop sync mode
Loop Length	10 - 4000 ms	500 ms	Loop duration (Free mode)
Mix	0 - 100%	100%	Dry/wet blend
Output	-20 to +20 dB	0 dB	Output level
Processing	Mono / Stereo	Stereo	Channel mode
Capture	Next / Last	Next	Grab the next or the last loop
Loop Speed	-200 to +200%	100%	Tape speed of the loop
Overdub	Off / On	Off	Record onto the loop
Feedback	0 - 100%	100%	Loop kept per Overdub pass
Resample	Button	Off	Print the sound into the loop
Warble	0 - 100%	0%	Tape wow
Loop Order	Stretch>Loop / Loop>Stretch	Stretch>Loop	Loop after or before the stretch

12. Tips & Tricks

11.1 Vintage Sampler Sound

- Set Sample Rate to 22050 Hz and Bit Depth to 12 for classic late-80s character
- Add subtle Preamp (10-20%) for warmth without obvious distortion
- Roll off highs with LP Filter around 12-15 kHz for that "sampled" quality

11.2 Filter Techniques

- High Resonance (70-100%): Creates acid-style squelchy filter sweeps
- Filter Drive with Low LP: Aggressive bass tones with controlled highs
- HP + LP together: Create bandpass effects by bringing both filters toward center

11.3 Creative Time Stretch

- Extreme slow (Time Factor 200-400%, Pitch Comp off): ambient pads from any source material
- Large grain (100-200ms) + Reverse: Granular textures with reversed grains
- Pitch Comp on: Time Factor becomes a pitch shift that keeps the timing
- Transpose: classic varispeed, pitch and speed move together

11.4 Lo-Fi Drums

- Bit Depth 8-10 bits adds crunch while maintaining punch
- Sample Rate around 16-22 kHz removes harsh transients naturally
- Use Mix at 50-70% to blend lo-fi character with clean attack

11.5 Gain Staging

- Enable Auto Gain when using heavy Preamp or Resonance settings
- Watch output levels when stacking Preamp + Filter Drive
- Use negative Output gain to compensate for boost from saturation

11.6 Printing with Resample

- Loop a bar (Capture Last), set Bit Depth to 6 bits, press Resample, then set Bit Depth back to 16: the loop stays crushed
- Loop Speed 50% and Resample: the octave-down loop is printed; press again for two octaves down
- Loop Order Loop>Stretch, Stretch on, Pitch Comp off, Time Factor 200%: press Resample, then switch Stretch off: the slowed loop stays
- Overdub on with Feedback around 70%: play into the loop and older layers fade on each pass

Support

For technical support, updates, and additional information:

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